



Spring 2026 MEMS 3430 Design of Thermal Systems

Design Homework 3: CCGT - Heat Exchanger Design

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Report Submission Date: April 7, 2026

I hereby certify that the lab report herein was completed in accordance with the McKelvey School of Engineering and Undergraduate Student academic integrity policies. I acknowledge that artificial intelligence (AI) tools were used to generate content included in this report.

Samantha Feldman

A handwritten signature in black ink that reads "Samantha Feldman".

Assumptions and Boundary Conditions

Assumptions

- The Brayton cycle operates with air and the Rankine cycle with water as working fluids, respectively.
- Possible damage of the turbine blades via erosion from droplet impingement should be minimized, so the turbine exit quality has a lower limit.
- Both the Brayton and Rankine cycles are non-ideal.
- Constant mass flow rate in each cycle (Rankine and Brayton).
- The heat exchanger that couples the Brayton and Rankine cycles consists of a preheater (economizer), evaporator, and superheater.
- The preheater brings the Rankine cycle working fluid from state 6 to saturation. It is a shell-and-tube heat exchanger with a single pass and 2 tube passes with liquid inside the tubes, and gas in the shell.
- The evaporator is a fire-tube design, hot exhaust gas from the Brayton cycle is in the tubes, and water from the Rankine cycle boils on the outside. Assume pool boiling is occurring. Assume the heat flux in the evaporator is well below critical heat flux (CHF).
- The superheater heats steam from the Rankine cycle from saturation to T_7 , it operates as a counterflow heat exchanger with 2 concentric tubes. Steam is in the outer annulus, and air flows inside the inner tube.
- Heat lost = heat gained from the Brayton to the Rankine for each stage of the heat exchanger.
- Air in the Brayton cycle is an ideal gas.
- No fouling occurs in the heat exchanger.
- Thermal resistance is negligible in phase change (boiling) compared to the convection of the single phase fluid (gas) as phase change typically has very high convective heat transfer coefficients.

- Mean temperature of a fluid between entering and exiting a portion of the heat exchanger (pre-heater, evaporator, superheater) is assumed constant and used when estimating thermophysical properties (e.g. viscosity, thermal conductivity, surface tension, ect).
- The 200 tubes inside the heat exchanger sections are thin walled.

Boundary Conditions

- The heat transfer between the Brayton and Rankine cycles is: $\dot{Q} = 30 \text{ MW}$.
- Number of tubes in heat exchanger: $N = 200$ tubes.
- Pre-heater single tube diameter: $d = 20 \text{ mm}$.
- Evaporator single tube diameter: $d = 50 \text{ mm}$.
- Superheater single tube diameters: outer diameter (d_o) = 80mm and inner diameter (d_i) = 50 mm.
- Brayton turbine isentropic efficiency (n_t) = 0.90, and Brayton compressor isentropic efficiency (n_c) = 0.80.
- Brayton cycle temperatures: $T_1 = 300K$, $T_3 = 1200K$, $T_4 = 650.99K$, $T_5 = 500K$.
- Brayton cycle optimal compression ratio $r = 12$.
- Brayton cycle mass flow rate: $\dot{m}_a = 197.90kg/s$.
- Rankine turbine isentropic efficiency (n_t) = 0.90.
- Rankine turbine exit quality lower limit: $x = 0.90$ [1].
- Rankine cycle temperatures: $T_9 = 16^\circ\text{C}$.
- Rankine cycle pressures: $P_8 = P_9 = 1.82kPa$, $P_6 = P_7 = 662kPa$.
- Rankine cycle mass flow rate: $\dot{m}_w = 9.58kg/s$.
- Heat exchanger tube layout: square pitch [2].

- Pitch to diameter ratio = 1.25 [3].
- Tube pitch for pre-heater: (center-to-center) = $1.25d_o = 0.025$ m [3].
- Minimum gap between adjacent tubes: 6.4 mm [3].
- Clearance between the bundle and shell inner wall: 13 mm. [4]
- Baffle spacing = $0.2 * D_s$ [5]
- Pipe roughness for commercial steel = 0.045 mm [6].

Diagrams

Figure 1 displays the schematic of a Combined Cycle Gas Turbine (CCGT). In this design homework, only the heat exchanger between cycles is being analyzed.

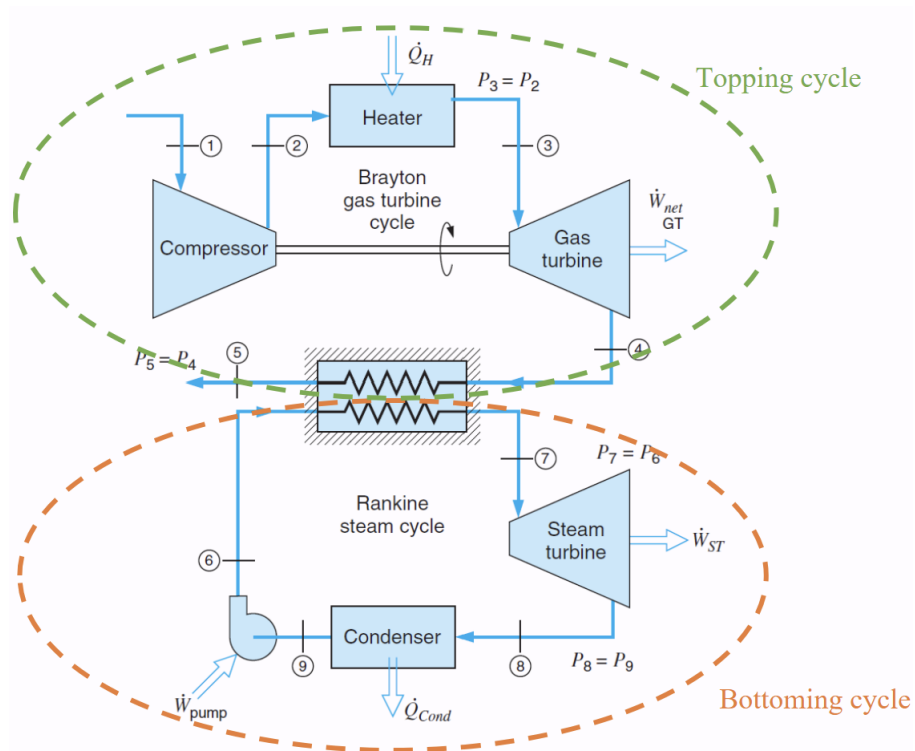


Figure 1 Schematic of a Combined Cycle Gas Turbine (CCGT).

Figure 2 displays the schematic of the succession of the three heat exchanger sections.

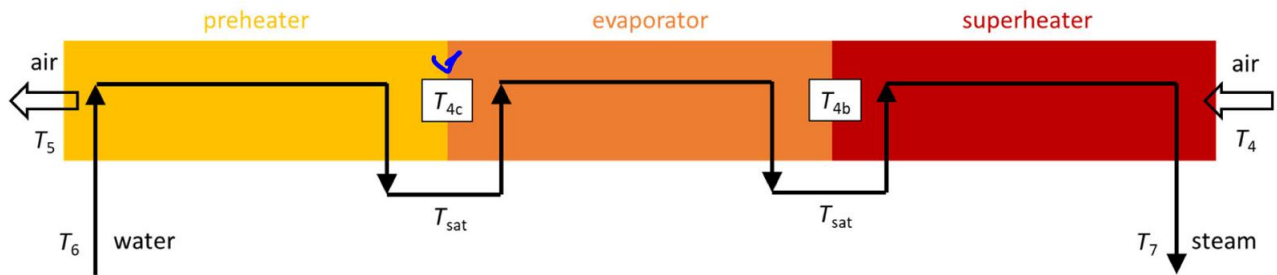


Figure 2 Schematic of the succession of the three heat exchanger sections.

Figure 3 shows the T-x diagram of the temperature profile in the three heat exchanger sections (pre-heater, evaporator, super-heater). The temperature numbers correspond to the temperature numbers displayed in Fig 2.

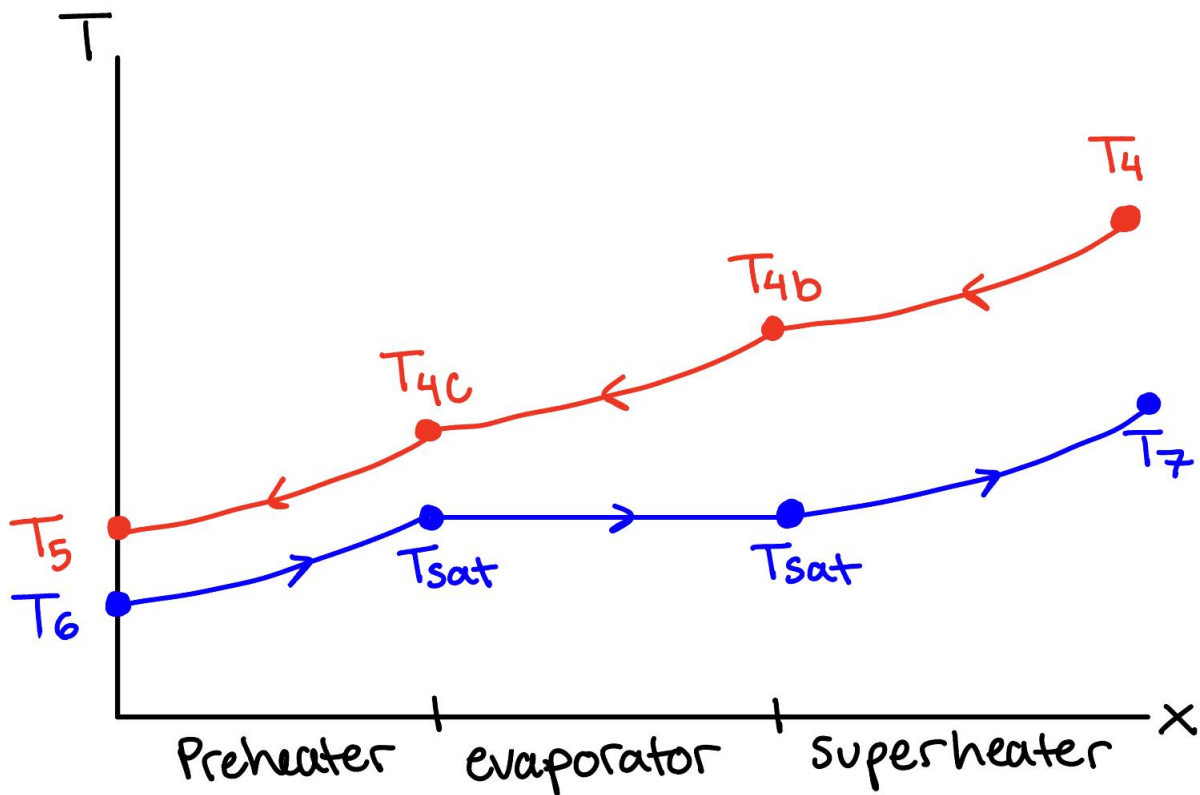


Figure 3 T-x diagram of the temperature profile of the 3 heat exchanger sections.

Equations

0.1 Temperature calculations. The following is the calculation of the transferred heat and inlet and outlet temperatures, respectively, of the three heat exchanger sections

0.1.1 *Pre-heater.* An equation for the heat transfer is given below that is used for the Brayton cycle.

$$\dot{Q} = \dot{m}c_p\Delta T \quad (1)$$

$$\Delta T = T_{4c} - T_5$$

Where $\dot{Q}$ is the rate of heat transfer [W], $\dot{m}$ is mass flow rate of the air [kg/s], c_p is specific heat capacity at constant pressure at the mean air temperature [kJ/kg-K], T_{4c} is the air temperature [K] entering the pre-heater, and T_5 is the air temperature exiting the heat exchanger.

A heat transfer equation that is used for the Rankine cycle is given below.

$$\dot{Q} = \dot{m}_w(h_o - h_i) \quad (2)$$

$$h_o = h_f(p_6) = 688 \text{ kJ/kg}$$

$$h_i = h_6 = 69.17 \text{ kJ/kg}$$

Where $\dot{m}$ is mass flow rate of the water [kg/s], h_o and h_i is the pre-heater exiting and entering enthalpy [kJ/kg] respectively. As the heat lost by the Brayton cycle is equivalent to the heat gained by the Rankine cycle, by setting equations 1 and 2 equal to each other the output T_{4c} can be isolated.

$$\dot{Q}_a = \dot{Q}_w = \dot{m}_a c_{p,a}(T_{4c} - T_5) = \dot{m}(h_o - h_i)$$

$$T_{4c} = \frac{\dot{m}_w(h_o - h_i)}{\dot{m}_a c_{p,a}} + T_5 \quad (3)$$

$$T_{4c} = 528.81 \text{ K}$$

Now inputting T_{4c} back into eq. 1 and checking with eq. 2 the heat transfer of the pre-heater is found to be shown below:

$$\dot{Q}_{pre} = 5.9MW$$

0.1.2 Evaporator. Using the same assumption that heat lost = heat gain between the Brayton and Rankine cycles with equations 1 and 2 the temperature T_{4b} can be calculated as such:

$$T_{4b} = \frac{\dot{m}_w(h_o - h_i)}{\dot{m}_a c_{p,a}} + T_{4c} \quad (4)$$

$$h_i = h_f(p_6) = 688kJ/kg$$

$$h_o = h_g(p_6) = 2760kJ/kg$$

$$T_{4b} = 625.25K$$

Where $\dot{m}_a$ is mass flow rate of the air [kg/s], $c_{p,a}$ is specific heat capacity at constant pressure at the mean air temperature [kJ/kg-K], T_{4c} is the air temperature [K] exiting the evaporator, and T_{4b} is the air temperature entering the evaporator. $\dot{m}_w$ is mass flow rate of the water [kg/s], h_o and h_i is the evaporator exiting and entering enthalpy [kJ/kg] respectively.

Now inputting T_{4b} back into eq. 1 and checking with eq. 2 the heat transfer of the evaporator is found to be shown below:

$$\dot{Q}_{evap} = 19.9MW$$

0.1.3 Superheater. Using the heat transfer calculated for both the pre-heater and evaporator, the heat transfer can be calculated for the superheater using the total heat transferred. It was also sanity checked by using the air temperatures to calculate the heat transfer independently with eq. 1.

$$\dot{Q}_{super} = \dot{Q}_{tot} - \dot{Q}_{evap} - \dot{Q}_{pre} \quad (5)$$

$$\dot{Q}_{super} = 4.2MW$$

0.2 Length calculations.

0.2.1 *Pre-heater.* To find the length for the shell and tube pre-heater, the guiding equations are given below.

$$NTU = \frac{UA}{C_{min}} \quad (6)$$

$$A_{pre} = \pi dNL_p \quad (7)$$

$$C = \dot{m}c_p \quad (8)$$

Where NTU is the number of transfer units, U is the overall heat transfer coefficient [W/m^2K], and C_{min} is the smaller of the two fluid heat capacity rates [W/K], $\dot{m}$ is the mass flow rate of the fluid, and c_p is the specific heat capacity at constant pressure [$J/kg * K$], and A is the surface area of the tubes. N is the number of tubes, d is the outer tube diameter [m], and L is the tube length [m]. These equations can be rearranged to provide an expression for the length of the pre-heater.

$$L_p = \frac{NTU * C_{min}}{U\pi dN} \quad (9)$$

To isolate and find U the following equation was used to describe the convection resistance.

$$\frac{1}{UA} = R_{c,c} + R_w + R_{c,h}$$

Where $R_{c,c}$, R_w , $R_{c,h}$ are the resistances [K/W] from the cold fluid convection, fouling, and hot fluid convection. This equation is further simplified by assuming there is no fouling to the equation below.

$$\frac{1}{U} = \frac{1}{h_a} + \frac{1}{h_w} \quad (10)$$

Where h_a and h_w are the convective heat transfer coefficients [W/m^2K] for air and water respectively. First looking at the air side of the pre-heater to find the h_a , the Reynolds number is computed with the equations below that take into account the baffles in the section.

$$Re_s = \frac{D_e G_s}{\mu} \quad (11)$$

$$G_s = \frac{\dot{m}_a}{S_m} \quad (12)$$

$$S_m = (bafflespacing)(S_D - d) \quad (13)$$

$$D_e = \frac{4(C_{p,d} S_d^2 - \frac{\pi d_o^2}{4})}{\pi d_o} \quad (14)$$

$$bafflespacing = 0.2D_s \quad (15)$$

$$D_s = \Delta + D_b \quad (16)$$

$$D_b = d_o * (N/K_1)^{1/n_1} \quad (17)$$

$$P_t = 1.25 * d \quad (18)$$

Where Re is the Reynold's number, D_e is the equivalent diameter [m], G_s is the shell-side mass flow rate [kg/s], μ is the viscosity [Pa-s] of the fluid. $C_{p,d}$ is 1 for the square pitch shell, S_D is the tube-tube pitch distance [m], d is the tube diameter [m], S_m is used to find the shell-side effective

crossflow area, and D_s is the shell diameter [m]. P_t is the tube pitch [m], K_1 and n_1 are constants from the standard tube-count tables for square pitch and 2 pass with values of 0.156 and 2.291 respectively [7], and Δ is the shell to bundle diametral clearance [m].

From there the Nussult (Nu) number needs to be evaluated. Seeing as the Reynold's number is turbulent, and the flow is external over a cylinder, the Nussult number equation is shown below.

$$Nu = CRe^m Pr^{1/3} \quad (19)$$

Where C and m are constants based on the magnitude of the Reynold's number, Pr is the Prandtl number, and Re is the Reynold's number. With all this information the shell side h_a value can be computed with the following equation.

$$h = \frac{Nuk}{d} \quad (20)$$

Where h is the convective heat transfer coefficient [W/m^2K] for the fluid, Nu is the Nussult number, k is the thermal conductivity [W/mK] of the fluid, and d is the diameter [m] of the tube.

Moving over to the tube side, where the water lives the heat transfer coefficient is calculated by first finding the Reynold's number (Re).

$$Re = \frac{4\dot{m}}{\mu\pi DN} \quad (21)$$

Where $\dot{m}$ is the mass flow rate of the water, μ is the viscosity of the water [Pa-s], D is the diameter of the tube, and N is the number of tubes. After calculating Re, and finding that it is 9700 for internal flow in a cylindrical tube, the corresponding Nu equation is below.

$$Nu = 0.023Re^{4/5} Pr^n \quad (22)$$

Where n is 0.4 because the water is being heated ($T_s > T_w$), Pr is the Prandtl number, and Re is the Reynold's number. The heat transfer coefficient for water can be calculated using eq. 22 and plugging it into eq. 20 by using the water coefficients that correspond to the mean temperature in the pre-heater.

Now that U is known, both shell and tube side C can be calculated with eq. 8, and compared to find which is bigger. With this knowledge the following information can be computed for a single shell with 2 tube passes:

$$C_r = \frac{C_{min}}{C_{max}} \quad (23)$$

$$NTU = -(1 + Cr^2)^{-1/2} \ln\left(\frac{E - 1}{E + 1}\right) \quad (24)$$

$$E = \frac{(2/\epsilon_1) - (1 + Cr)}{(1 + Cr^2)^{1/2}} \quad (25)$$

Where C is the maximum or minimum fluid heat capacity rate [W/K] and ϵ is the heat exchanger effectiveness. After computing these values, they can all be plugged into eq. 9 to find the length of the pre-heater.

0.2.2 Evaporator. The Reynold's number for air flow inside the tube can be calculated with eq. 21 and found to be $1.87 * 10^6$ the following Nussult number equation applies.

$$Nu = 0.023 * Re^{4/5} * Pr^n \quad (26)$$

Where n is 0.3 for cooling of the fluid. Using this equation the the convective heat transfer coefficient [W/m^2K] h can be found, as with the assumption that thermal resistance is negligible for single phase flows, h_w is essentially zero and the following equation applies.

$$U = h_a \quad (27)$$

$$h_a = \frac{Nuk_a}{d} \quad (28)$$

h_a can be found using eq. 28, and U is the overall heat transfer coefficient [W/m^2K] for the flow in the evaporator.

Finding the length of the evaporator section requires the log-mean method. With the following equations.

$$\dot{Q} = UA\Delta T_{lm} \quad (29)$$

$$T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)} \quad (30)$$

$$\Delta T_1 = T_i - T_{sat}$$

$$\Delta T_2 = T_o - T_{sat}$$

Where A is the surface area [m^2] of the evaporator, U is the overall heat transfer coefficient [W/m^2K], and ΔT_{lm} is the log mean temperature difference, T_o and T_i are the temperatures [K] of the air leaving and entering the evaporator, respectively, and T_{sat} is the saturated temperature [K] of the water in the evaporator. The final length equation for the evaporator is thus.

$$L_E = \frac{\dot{Q}_E}{\Delta T_{lm} h \pi d N} \quad (31)$$

0.2.3 Superheater. For the air side in the inner tube area, the specific heat capacities were calculated with eq. 8 to determine which was the max and the min. The Reynold's number was calculated using eq. 21 and found to be turbulent. This led to finding the Nussult equation which is given below.

$$Nu = 0.023 * Re^{4/5} * Pr^n \quad (32)$$

When calculating Nu for the air side $n = 0.3$ because it is cooling down, and for the steam side $n = 0.4$ because the fluid is heating up. This was then used to evaluate the heat transfer coefficient with eq. 20 for both the steam and air fluid sides. The effectiveness and NTU were then found with the following cross flow equations.

$$\epsilon = \frac{T_{w,o} - T_{w,i}}{T_{a,i} - T_{w,i}} \quad (33)$$

$$NTU = \frac{1}{C_r - 1} \ln\left(\frac{\epsilon - 1}{\epsilon C_r - 1}\right) \quad (34)$$

Where $T_{w,o}$, $T_{w,i}$, and $T_{a,i}$ is the temperature of the water exiting the super-heater, water entering the superheater, and air entering the superheater. ϵ is the heat exchanger section effectiveness. C_r is calculated using eq. 23.

Looking at the water inside the annulus the Reynolds number was calculated using the equation below.

$$Re = \frac{\rho u D_h}{\mu} \quad (35)$$

$$u = \frac{\dot{m}}{\rho A_c} \quad (36)$$

$$A_c = \frac{\pi(D_o^2 - D_i^2)}{4} \quad (37)$$

Where D_o and D_i are the outer and inner diameters [m] of the superheater tubes. Nu is able to be calculated using eq. 32, h_a and h_w are calculated with eq. 20, and U is calculated with 10. Finally the length of the super-heater was calculated with the equation below.

$$L_S = \frac{NTU * C_{min}}{U \pi d N} \quad (38)$$

0.3 Heat Flux calculations. To find the actual heat flux [W/m^2] in the evaporator the following equation is used.

$$q''_{actual} = \dot{Q}/A \quad (39)$$

$$A = \pi D L N \quad (40)$$

Where $\dot{Q}$ is the heat transfer [W] in the evaporator between the Brayton and Rankine cycles, A is the surface area [m^2] of the tubes, D is the diameter [m] of the tube, L is the length [m] of the evaporator, and N is the number of tubes.

The critical heat flux (CHF) [W/m^2] serves as a very high limit to the actual heat flux is below.

$$CHF = C_z h_{fg} \rho_v \left[\frac{\sigma_{lv} g (\rho_l - \rho_v)}{\rho_v^2} \right]^{1/4} \quad (41)$$

Where C_z is the zuber constant and is 0.131 for cylinders, h_{fg} is the latent heat of vaporization [kJ/kg], ρ_v is the density [kg/m^3] of the vapor, ρ_l is the density [kg/m^3] of the liquid, g is the gravity constant (9.81 m/s^2), and σ_{lv} is the surface tension [N/m].

RESULTS

Table 1 is a recap of the most important temperatures and mass flow parameters for the Rankine and Brayton cycles. All of the subscript numbers correspond to the state numbers in fig. 1.

Table 1 Re-cap of most important values from Brayton and Rankine cycles

Value	Quantity
T_1	300 K
T_3	1200 K
T_4	651 K
T_5	500 K
Brayton cycle mass flow rate $\dot{m}_a$	197.89 kg/s
T_6	289 K
T_7	639 K
T_9	289 K
Rankine cycle mass flow rate $\dot{m}_w$	9.58 kg/s
Saturation temperature in the heat exchanger (T_{sat})	436 K

Table 2 lists the values at the optimized layout of the heat exchanger.

Table 2 Heat Exchanger Values

Value	Quantity
T_{4b}	625.25 K
T_{4c}	528.81 K
Pre-heater Heat Transfer Rate $\dot{Q}_{pre}$	5.93 MW
Evaporator Heat Transfer Rate $\dot{Q}_{evap}$	19.85 MW
Super-heater Heat Transfer Rate $\dot{Q}_{sup}$	4.22 MW
Pitch Configuration	Square
Pitch Distance	0.025 m
Baffle Spacing	0.1 m
Shell Diameter	0.5 m
Pre-heater Length	2.37 m
Evaporator Length	4.9 m
Super-heater Length	7.32 m
Actual heat flux	0.13 MW/m^2
Critical heat flux	2.37 MW/m^2

Discussion and Sanity Check

The heat exchanger configuration presented in Table 2 shows a consistent temperature progression through the three sections. The intermediate gas temperatures decrease from the Brayton exhaust at $T_4 = 651$ K, through $T_{4b} = 625.25$ K after the super-heater, and further to $T_{4c} = 528.81$ K after the evaporator, before finally exhausting toward the pre-heater exit. All of these intermediate gas temperatures remain comfortably above the saturation temperature of $T_{sat} = 436$ K, ensuring a positive temperature difference is kept across every section of the heat exchanger, which is a thermodynamic requirement. The heat distribution further show heat exchanger is physically consistent: the evaporator is the largest at 19.85 MW (about 66% of the 30 MW total), while the pre-heater and super-heater contribute 5.93 MW and 4.22 MW respectively. This split is entirely expected, as the latent heat of

vaporization of water is large compared to the heating required in either the pre-heating or superheating stages.

The geometric and thermal sizing results also pass a physical sanity check. The square pitch configuration with a pitch distance of 0.025 m and baffle spacing of 0.1 m within a 0.5 m shell diameter are all standard and practical values for a shell-and-tube heat exchanger of this scale. The section lengths of 2.37 m, 4.90 m, and 7.32 m for the pre-heater, evaporator, and super-heater respectively make sense. While carrying the smallest heat transfer, the super-heater is the longest section. This is physically understandable by the lower convective heat transfer coefficient associated with single-phase gas-to-superheated-steam heat exchange compared to the boiling regime in the evaporator, which benefits from the additional heat transfer of boiling. Most importantly, the actual heat flux of 0.13 MW/m^2 is more than an order of magnitude below the critical heat flux of 2.37 MW/m^2 , giving a safety margin of about 18×. This confirms that the evaporator operates well in the boiling regime and that there is no risk of burnout or film boiling, validating the thermal design of the heat exchanger.

References

- [1] Murugan, R. S. and Subbarao, P. V., 2008, "Thermodynamic Analysis of Rankine-Kalina Combined Cycle," Int. J. of Thermodynamics, accessed March 1, 2026, <https://dergipark.org.tr/tr/download/article-file/65735>
- [2] Subramanian, R. S., 2003, "Shell-and-Tube Heat Exchangers," Department of Chemical and Biomolecular Engineering Clarkson University, accessed March 31, 2026, <https://lin-web.clarkson.edu/projects/subramanian/ch330/notes/Shell%20and%20Tube%20Heat%20Exchangers.pdf>
- [3] Nusantara, B. P., ed., 2006, "How to Calculate and Design Shell and Tube-type Heat Exchanger with a Single Heat Transfer," ASEAN Journal for Science and Engineering, accessed March 31, 2026, <https://ejournal.bumipublikasinusantara.id/index.php/ajsem/article/view/400>
- [4] Towler, G. and Sinnott, R., 2013, *Coulson & Richardson's Chemical Engineering, Volume 6: Chemical Engineering Design*, 5th ed., Elsevier, See Figure 12.10 for shell-bundle clearance vs. bundle diameter.
- [5] "The Effect of a Number of Baffles on the Performance of Shell-and-Tube Heat Exchangers," .
- [6] 2026, "Pipe Roughness," pipeflow.com, accessed March 31, 2026, <https://www.pipeflow.com/pipe-pressure-drop-calculations/pipe-roughness>
- [7] Towler, G. and Sinnott, R., 2013, *Coulson & Richardson's Chemical Engineering, Volume 6: Chemical Engineering Design*, 5th ed., Elsevier, See Table 12.4 for constants for bundle diameter equation (K_1 and n_1 for various tube passes and pitch types).

A Acknowledgment of Artificial Intelligence Resources

AI tools were used to look up LaTeX and Matlab code and was verified by inputting into Overleaf and Matlab respectively. Additionally, AI tools were used as a starting point for literature research, all information included from the output was verified by consulting the source. Anything not verified was not included in the report. A list of the AI prompts used is included below.

Cited AI Prompts

- ChatGPT, response to “between the mojave river, colorado river, lake mead, and lake havasu which would be the best coolant for a rankine cycle condensor for solar power plant in mojave dessert” OpenAI, February 28, 2026.
- ChatGPT, response to ”provide approximate intake temperature for colorado river cold water and lake mead from deep intake” OpenAI, February 28, 2026.
- Gemini, response to “pinch point range for steam turbine industrial rankine cycle for heat exchanger” Google, February 29, 2026.
- Copilot, response to “Take existing Matlab code and have it print these values: wt, wp at $x = 0.9$ ” Microsoft, March 3, 2026.
- Copilot, response to “Why is this latex code throwing an error” Microsoft, March 5, 2026.

B MATLAB Code

DHW3 Code

```
1 %% Combined Brayton-Rankine Cycle Heat Exchanger Calculations
2 % Covers: Pre-heater, Evaporator, Superheater
3 % Sections: Temperature calcs, Length calcs, Heat Flux calcs
4 % =====
5 % Temperature reference:
6 %   Pre-heater:   Air in=514K, Air out=500K, Water in=289K, Water out=436K
7 %   Evaporator:   Air in=T4b=562K, Air out=514K, Water=436K (phase change)
8 %   Superheater:   Air in=573K(T4), Air out=T4b, Water in=436K, Water out=561K
9 % =====
10
11 clear; clc;
12
13 %% =====
14 % GLOBAL INPUTS
15 % =====
16 m_dot_a   = 197.89;      % [kg/s]   Air mass flow rate (Brayton side)
17 m_dot_w   = 9.58;       % [kg/s]   Water mass flow rate (Rankine side)
18 N         = 200;        % [-]      Number of tubes
19 S_D       = 0.025;      % [m]      Tube-to-tube pitch distance
20 Q_dot_tot = 30e6;       % [W]      Total heat transferred
21
22 %% =====
23 % SECTION 1: PRE-HEATER
24 % =====
25 fprintf('=====\\n');
26 fprintf('          PRE-HEATER CALCULATIONS\\n');
27 fprintf('=====\\n\\n');
28
29 % --- Geometry ---
30 d_pre = 0.020;          % [m] tube diameter
31
32 % --- Enthalpies (Rankine side) ---
33 h_i_pre = 69.17;        % [kJ/kg] water ENTERING pre-heater (compressed liquid)
```

```

34 h_o_pre = 688;           % [kJ/kg] water EXITING pre-heater (sat. liquid at p6)
35
36 % --- Known temperatures ---
37 T5           = 500;       % [K] air EXITING pre-heater
38 T_w_in_pre = 289;         % [K] water ENTERING pre-heater (cold side inlet)
39                 % corresponds to h_i = 69.17 kJ/kg
40
41 % --- Air cp ---
42 cp_a_pre     = 1.04;      % [kJ/kg-K]
43 cp_a_pre_SI  = cp_a_pre * 1000; % [J/kg-K]
44
45 % --- Q from water side energy balance ---
46 Q_dot_pre = m_dot_w * (h_o_pre - h_i_pre) * 1000; % [W]
47
48 % --- T4c: air temperature ENTERING pre-heater ---
49 T4c = Q_dot_pre / (m_dot_a * cp_a_pre_SI) + T5;
50
51 fprintf('--- Temperature Results ---\n');
52 fprintf('T5 (air out):           %.2f K (%.1f C)\n', T5, T5-273.15);
53 fprintf('T4c (air in):           %.2f K (%.1f C)\n', T4c, T4c-273.15);
54 fprintf('T_w_in (water in):       %.2f K (%.1f C)\n', T_w_in_pre, ...
55         T_w_in_pre-273.15);
56
57 fprintf('Q_dot_pre:                %.4f MW\n\n', Q_dot_pre/1e6);
58
59 % --- Air (shell side) fluid properties at mean air temp ~507 K ---
60 mu_a_pre = 270.1e-7;      % [Pa-s]
61 k_a_pre  = 40.7e-3;       % [W/m-K]
62 Pr_a_pre = 0.684;         % [-]
63
64 % --- Water (tube side) fluid properties at mean water temp ~363 K ---
65 mu_w_pre = 324e-6;       % [Pa-s]
66 k_w_pre  = 674e-3;       % [W/m-K]
67 Pr_w_pre = 2.02;         % [-]
68 cp_w_pre = 4.204e3;      % [J/kg-K]
69
70 % --- Nusselt constants ---
71 C_shell = 0.193;         % shell side (external flow, Re range ~2e4-1e5)
72 m_shell = 0.618;

```

```

71 n_pre    = 0.4;                % tube side heating exponent
72
73 % --- Shell geometry ---
74 Ds_pre    = 0.5;
75 baffle_pre = 0.2 * Ds_pre;
76 Sm_pre    = baffle_pre * (S.D - d_pre);
77 Cp_d      = 1;                % square pitch
78 De_pre    = 4 * (Cp_d*S.D^2 - pi*d_pre^2/4) / (pi*d_pre);
79 Gs_pre    = m_dot_a / Sm_pre;
80 Re_s_pre  = De_pre * Gs_pre / mu_a_pre;
81
82 Nu_a_pre  = C_shell * Re_s_pre^m_shell * Pr_a_pre^(1/3);
83 h_a_pre   = Nu_a_pre * k_a_pre / d_pre;
84
85 % --- Tube side (water) ---
86 Re_w_pre  = 4 * m_dot_w / (mu_w_pre * pi * d_pre * N);
87 Nu_w_pre  = 0.023 * Re_w_pre^(4/5) * Pr_w_pre^(n_pre);
88 h_w_pre   = Nu_w_pre * k_w_pre / d_pre;
89
90 % --- Overall U (no fouling) ---
91 U_pre     = 1 / (1/h_a_pre + 1/h_w_pre);
92
93 % --- Capacity rates ---
94 C_a_pre   = m_dot_a * cp_a_pre_SI;
95 C_w_pre   = m_dot_w * cp_w_pre;
96 C_min_pre = min(C_a_pre, C_w_pre);
97 C_max_pre = max(C_a_pre, C_w_pre);
98 Cr_pre    = C_min_pre / C_max_pre;
99
100 % --- Effectiveness ---
101 % Q_max = C_min * (T_hot_in - T_cold_in)
102 % Hot inlet = T4c (air entering pre-heater)
103 % Cold inlet = T_w_in_pre (water entering pre-heater)
104 Q_max_pre = C_min_pre * (T4c - T_w_in_pre);
105 eps_pre   = Q_dot_pre / Q_max_pre;
106
107 fprintf('--- Pre-heater Length Results ---\n');
108 fprintf('Re-s (shell/air):    %.2f\n',      Re_s_pre);

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109 fprintf('Re (tube/water): %.2f\n', Re_w_pre);
110 fprintf('h_a: %.2f W/m^2K\n', h_a_pre);
111 fprintf('h_w: %.2f W/m^2K\n', h_w_pre);
112 fprintf('U: %.2f W/m^2K\n', U_pre);
113 fprintf('C_air: %.2f W/K\n', C_a_pre);
114 fprintf('C_water: %.2f W/K\n', C_w_pre);
115 fprintf('Cr: %.4f\n', Cr_pre);
116 fprintf('Effectiveness: %.4f\n', eps_pre);
117
118 if eps_pre ≥ 1
119     warning('eps_pre = %.4f ≥ 1. Check temperatures or Q values.', eps_pre);
120 end
121
122 % --- NTU (1 shell, 2 tube passes) ---
123 E_pre = (2/eps_pre - (1 + Cr_pre)) / sqrt(1 + Cr_pre^2);
124 NTU_pre = -(1/sqrt(1 + Cr_pre^2)) * log((E_pre - 1)/(E_pre + 1));
125
126 % --- Length ---
127 L_pre = (NTU_pre * C_min_pre) / (U_pre * pi * d_pre * N);
128
129 fprintf('NTU: %.4f\n', NTU_pre);
130 fprintf('Pre-heater Length: %.4f m\n\n', L_pre);
131
132 %% =====
133 % SECTION 2: EVAPORATOR
134 % =====
135 fprintf('=====\\n');
136 fprintf(' EVAPORATOR CALCULATIONS\\n');
137 fprintf('=====\\n\\n');
138
139 % --- Geometry ---
140 d_evap = 0.05; % [m] tube diameter
141
142 % --- Enthalpies (Rankine side) ---
143 h_i_evap = 688; % [kJ/kg] water ENTERING evaporator = h_f(p6), sat. ...
    liquid
144 h_o_evap = 2760; % [kJ/kg] water EXITING evaporator = h_g(p6), sat. vapor
145

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146 % --- Air cp ---
147 cp_a_evap = 1.04; % [kJ/kg-K]
148 cp_a_evap_SI = cp_a_evap * 1000;
149
150 % --- T4b: air temperature ENTERING evaporator ---
151 T4b = (m_dot_w * (h_o_evap - h_i_evap)) / (m_dot_a * cp_a_evap) + T4c;
152
153 % --- Q_evap from air side ---
154 Q_dot_evap = m_dot_a * cp_a_evap_SI * (T4b - T4c); % [W]
155
156 fprintf('--- Temperature Results ---\n');
157 fprintf('T4b (air in): %.2f K (%.1f C)\n', T4b, T4b-273.15);
158 fprintf('T4c (air out): %.2f K (%.1f C)\n', T4c, T4c-273.15);
159 fprintf('T_sat (water): 436.00 K (162.9 C)\n');
160 fprintf('Q_dot_evap: %.4f MW\n\n', Q_dot_evap/1e6);
161
162 % --- Air properties at mean evaporator air temp ~538 K ---
163 mu_a_evap = 280e-7; % [Pa-s]
164 k_a_evap = 0.04; % [W/m-K]
165 Pr_a_evap = 0.683; % [-]
166 T_sat = 436; % [K] saturation temp of water in evaporator
167 n_evap = 0.3; % Dittus-Boelter: cooling of air
168
169 % --- Re for air INSIDE tube ---
170 Re_a_evap = 4 * m_dot_a / (mu_a_evap * pi * d_evap * N);
171
172 % --- Nu and h_a ---
173 Nu_a_evap = 0.023 * Re_a_evap^(4/5) * Pr_a_evap^n_evap;
174 h_a_evap = Nu_a_evap * k_a_evap / d_evap;
175
176 % --- Phase change: U = h_a (water-side resistance negligible) ---
177 U_evap = h_a_evap;
178
179 % --- LMTD ---
180 % Air flows inside tubes (hot), water boils outside (constant T_sat)
181 % dT1 = T_air_in - T_sat (hot end)
182 % dT2 = T_air_out - T_sat (cold end)
183 dT1_evap = T4b - T_sat;

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184 dT2_evap = T4c - T_sat;
185 T_lm_evap = (dT1_evap - dT2_evap) / log(dT1_evap / dT2_evap);
186
187 % --- Length ---
188 %L_evap = (m_dot_a * cp_a_evap_SI * log(dT1_evap/dT2_evap)) / ...
189           %(h_a_evap * pi * d_evap * N);
190 L_evap = (Q_dot_evap)/(T_lm_evap*h_a_evap * pi * d_evap * N);
191
192 fprintf('--- Evaporator Length Results ---\n');
193 fprintf('Re_a (air):           %.4e\n',      Re_a_evap);
194 fprintf('Nu_a:                %.2f\n',      Nu_a_evap);
195 fprintf('h_a:                  %.2f W/m^2K\n', h_a_evap);
196 fprintf('U (= h_a):              %.2f W/m^2K\n', U_evap);
197 fprintf('T_lm:                    %.2f K\n',      T_lm_evap);
198 fprintf('Evaporator Length:      %.4f m\n\n',      L_evap);
199
200 %% =====
201 % SECTION 3: SUPERHEATER
202 % =====
203 fprintf('===== \n');
204 fprintf('          SUPERHEATER CALCULATIONS \n');
205 fprintf('===== \n\n');
206
207
208 % --- Air (inner tube) properties at mean air temp ~568 K ---
209 mu_a_sup = 288.4e-7;      % [Pa-s]
210 k_a_sup = 43.9e-3;        % [W/m-K]
211 Pr_a_sup = 0.683;         % [-]
212 cp_a_sup = 1.04e3;        % [J/kg-K]
213
214 % --- Steam (annulus) properties at mean steam temp ~500 K ---
215 mu_w_sup = 16.69e-6;      % [Pa-s]
216 k_w_sup = 42.3e-3;        % [W/m-K]
217 Pr_w_sup = 1.28;          % [-]
218 cp_w_sup = 3.27e3;        % [J/kg-K]
219 rho_w_sup = 1/0.0766;     % [kg/m^3]
220
221 % --- Superheater geometry ---

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222 D_o_sup = 0.08;           % [m] outer annulus diameter
223 D_i_sup = 0.05;           % [m] inner tube diameter
224
225 % --- Known temperatures ---
226 % Air:   T_a_in = T4 = 650.99 K,   T_a_out = T4b (calculated above)
227 % Steam: T_w_in = 436 K (sat),   T_w_out = 561 K
228 T_a_i_sup = 650.99;         % [K] air ENTERING superheater = T4
229 T_a_o_sup = T4b;            % [K] air EXITING  superheater = T4b
230 T_w_i_sup = 436;            % [K] steam ENTERING superheater (sat. vapor)
231 T_w_o_sup = 561;            % [K] steam EXITING  superheater
232
233 % --- Q_super from energy balance ---
234 Q_dot_super = Q_dot_tot - Q_dot_evap - Q_dot_pre;
235 %Q_dot_super = m_dot_a * cp_a_sup * (T_a_i_sup - T4b);
236
237 fprintf('--- Heat Results ---\n');
238 fprintf('Q_dot_super:           %.4f MW\n\n', Q_dot_super/1e6);
239
240 % Sanity check: steam outlet must be below air inlet
241 if T_w_o_sup >= T_a_i_sup
242     warning('T_w_o_sup (%.1f K) >= T_a_i_sup (%.1f K): thermodynamically ...
243             impossible.', ...
244             T_w_o_sup, T_a_i_sup);
245 end
246
247 fprintf('--- Temperature Check ---\n');
248 fprintf('T_a_in   (air in  = T4):   %.2f K\n', T_a_i_sup);
249 fprintf('T_a_out   (air out = T4b):  %.2f K\n', T_a_o_sup);
250 fprintf('T_w_in   (steam in):         %.2f K\n', T_w_i_sup);
251 fprintf('T_w_out   (steam out):         %.2f K\n\n', T_w_o_sup);
252
253 % --- Air side Re, Nu, h (inside tube) ---
254 Re_a_sup = 4 * m_dot_a / (mu_a_sup * pi * d_evap * N);
255 n_sup     = 0.4;           % Dittus-Boelter: heating of steam
256 Nu_a_sup = 0.023 * Re_a_sup^(4/5) * Pr_a_sup^(0.3);
257 h_a_sup  = Nu_a_sup * k_a_sup / d_evap;
258
259 % --- Steam side Re, Nu, h (annulus) ---

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```

259 A_c_sup = pi * (D_o_sup^2 - D_i_sup^2) / 4; % annulus cross-section
260 D_h_sup = D_o_sup - D_i_sup; % hydraulic diameter
261 u_sup = m_dot_w / (rho_w_sup * A_c_sup);
262 Re_w_sup = rho_w_sup * u_sup * D_h_sup / mu_w_sup;
263 Nu_w_sup = 0.023 * Re_w_sup^(4/5) * Pr_w_sup^(0.4); %CHANGE BACK TO SAME AS ...
    AIR, JUST FIX N VALUE TO PR NUM 0.4 FOR STEAM AND 0.3 FOR AIR
264 h_w_sup = Nu_w_sup * k_w_sup / D_h_sup;
265
266 % Velocity sanity check
267 rho_a_sup = 1.04e3 / (287 * T_a_i_sup); % ideal gas, approximate
268 A_tube_sup = N * pi * d_evap^2 / 4;
269 u_a_sup = m_dot_a / (rho_a_sup * A_tube_sup);
270 fprintf('Air velocity in superheater tubes: %.1f m/s\n', u_a_sup);
271
272 % --- Overall U ---
273 U_sup = 1 / (1/h_a_sup + 1/h_w_sup);
274
275 % --- Capacity rates ---
276 C_a_sup = m_dot_a * cp_a_sup;
277 C_w_sup = m_dot_w * cp_w_sup;
278 C_min_sup = min(C_a_sup, C_w_sup);
279 C_max_sup = max(C_a_sup, C_w_sup);
280 Cr_sup = C_min_sup / C_max_sup;
281
282 % --- Effectiveness (based on steam side water is C_min) ---
283 % eps = (T_w_out - T_w_in) / (T_a_in - T_w_in)
284 eps_sup = (T_w_o_sup - T_w_i_sup) / (T_a_i_sup - T_w_i_sup);
285
286 if eps_sup >= 1
287     warning('eps_sup = %.4f > 1. Check superheater temperatures.', eps_sup);
288 end
289
290 % --- NTU (counter-flow, which is appropriate for double-pipe) ---
291 % Old formula: NTU = 1/(Cr-1) * ln((eps-1)/(eps*Cr-1))
292 % Counter-flow, correct formula:
293 NTU_sup = (1/(1 - Cr_sup)) * log((1 - eps_sup*Cr_sup) / (1 - eps_sup));
294
295 % Pre-heater (1 shell, 2 tube passes), your E method is correct in form,

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296 % but verify the sign convention matches your textbook exactly.
297
298 lm_sup = ((T_a_i_sup-T_w_o_sup) - (T_a_o_sup - ...
           T_w_i_sup))/log((T_a_i_sup-T_w_o_sup)/(T_a_o_sup - T_w_i_sup));
299 % --- Length ---
300 %L_sup = (NTU_sup * C_min_sup) / (pi * d_evap * U_sup * N);
301 L_sup = Q_dot_super/(N*U_sup*d_evap*lm_sup*0.15*pi);
302
303 fprintf('--- Superheater Length Results ---\n');
304 fprintf('Re_a (air, inner): %.4e\n',      Re_a_sup);
305 fprintf('Re_w (steam, annul): %.4e\n',      Re_w_sup);
306 fprintf('h_a: %.2f W/m^2K\n', h_a_sup);
307 fprintf('h_w: %.2f W/m^2K\n', h_w_sup);
308 fprintf('U: %.2f W/m^2K\n', U_sup);
309 fprintf('C_air: %.2f W/K\n', C_a_sup);
310 fprintf('C_steam: %.2f W/K\n', C_w_sup);
311 fprintf('Cr: %.4f\n', Cr_sup);
312 fprintf('Effectiveness: %.4f\n', eps_sup);
313 fprintf('NTU: %.4f\n', NTU_sup);
314 fprintf('Superheater Length: %.4f m\n\n', L_sup);
315
316 %% =====
317 % SECTION 4: HEAT FLUX (Evaporator)
318 % =====
319 fprintf('===== \n');
320 fprintf('          HEAT FLUX CALCULATIONS (Evaporator) \n');
321 fprintf('===== \n\n');
322
323 % --- Actual heat flux ---
324 A_evap_flux = pi * d_evap * L_evap * N;          % [m^2]
325 q_actual    = Q_dot_evap / A_evap_flux;          % [W/m^2]
326
327 % --- Critical Heat Flux (Zuber correlation for cylinders) ---
328 C_z          = 0.131;          % [-]      Zuber constant
329 h_fg          = 2070e3;          % [J/kg]   Latent heat of vaporization at T_sat
330 rho_v         = 1/0.261;          % [kg/m^3] Vapor density
331 rho_l         = 1/1.11e-3;          % [kg/m^3] Liquid density
332 sigma_lv      = 0.0451;          % [N/m]    Surface tension at T_sat

```

```

333 g          = 9.81;          % [m/s^2]
334
335 CHF = C_z * h_fg * rho_v * ...
336         (sigma_lv * g * (rho_l - rho_v) / rho_v^2)^(1/4);
337
338 fprintf('--- Heat Flux Results ---\n');
339 fprintf('Evaporator surface area: %.4f m^2\n',          A_evap_flux);
340 fprintf('Actual heat flux:          %.2f W/m^2   (%.4f MW/m^2)\n', q_actual, ...
        q_actual/1e6);
341 fprintf('Critical heat flux:        %.2f W/m^2   (%.4f MW/m^2)\n', CHF,          ...
        CHF/1e6);
342 if q_actual < CHF
343     fprintf('STATUS: Actual flux is BELOW critical heat flux. OK.\n');
344 else
345     fprintf('WARNING: Actual flux EXCEEDS critical heat flux!\n');
346 end
347
348 %% =====
349 %  SUMMARY
350 % =====
351 fprintf('\n===== \n');
352 fprintf('                SUMMARY\n');
353 fprintf('===== \n');
354 fprintf('--- Temperatures ---\n');
355 fprintf('T5   (air out pre-heater):      %.2f K   (%.1f C)\n', T5,   T5-273.15);
356 fprintf('T4c  (air in  pre-heater):      %.2f K   (%.1f C)\n', T4c, T4c-273.15);
357 fprintf('T4b  (air in  evaporator):        %.2f K   (%.1f C)\n', T4b, T4b-273.15);
358 fprintf('T4   (air in  superheater):        %.2f K   (%.1f C)\n', T_a_i_sup, ...
        T_a_i_sup-273.15);
359 fprintf('--- Heat Transfer ---\n');
360 fprintf('Q_pre:                          %.4f MW\n', Q_dot_pre/1e6);
361 fprintf('Q_evap:                          %.4f MW\n', Q_dot_evap/1e6);
362 fprintf('Q_super:                         %.4f MW\n', Q_dot_super/1e6);
363 fprintf('Q_total:                         %.4f MW\n', ...
        (Q_dot_pre+Q_dot_evap+Q_dot_super)/1e6);
364 fprintf('--- Lengths ---\n');
365 fprintf('L_pre:                            %.4f m\n', L_pre);
366 fprintf('L_evap:                           %.4f m\n', L_evap);

```

```
367 fprintf('Lsuper: %.4f m\n', Lsup);
```